

NACS 645 – The need for paradigms

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DEPARTMENT OF
PSYCHOLOGY

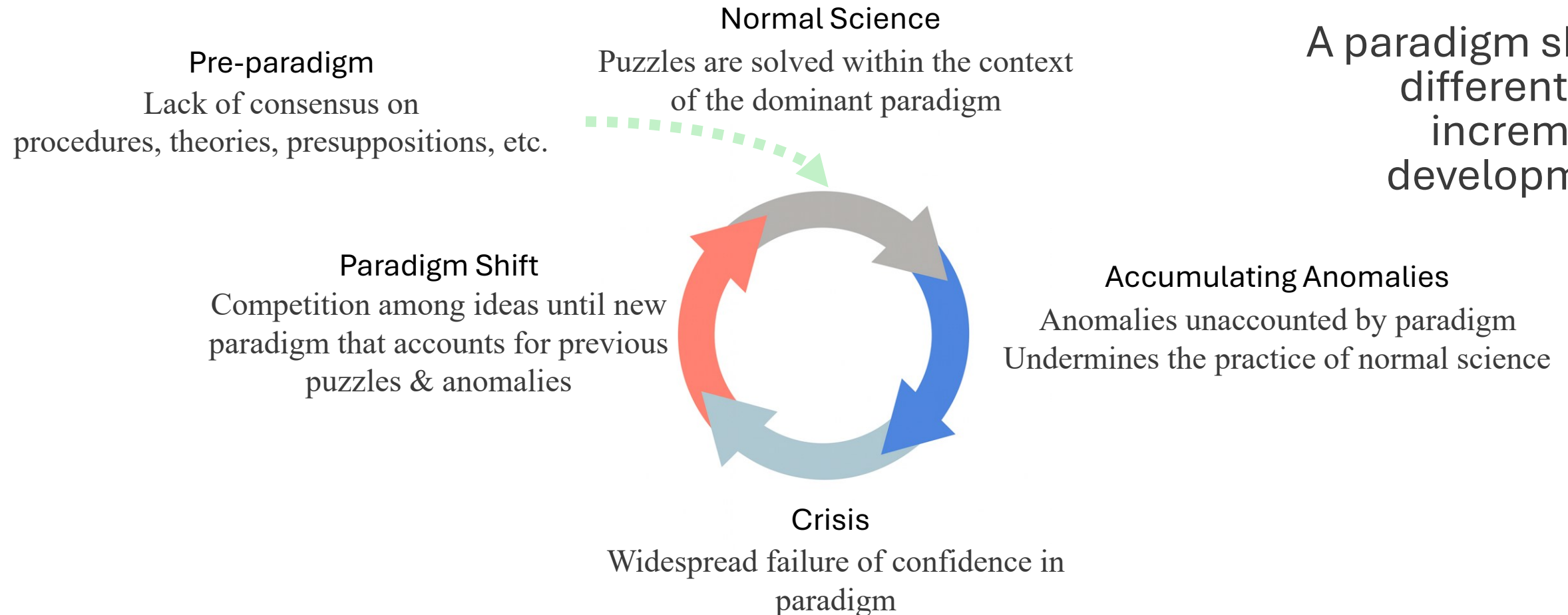


PROGRAM IN
NEUROSCIENCE &
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Scientific revolutions

A paradigm can be defined as the generally accepted concepts and practices of a field.

A paradigm shift is different from incremental developments



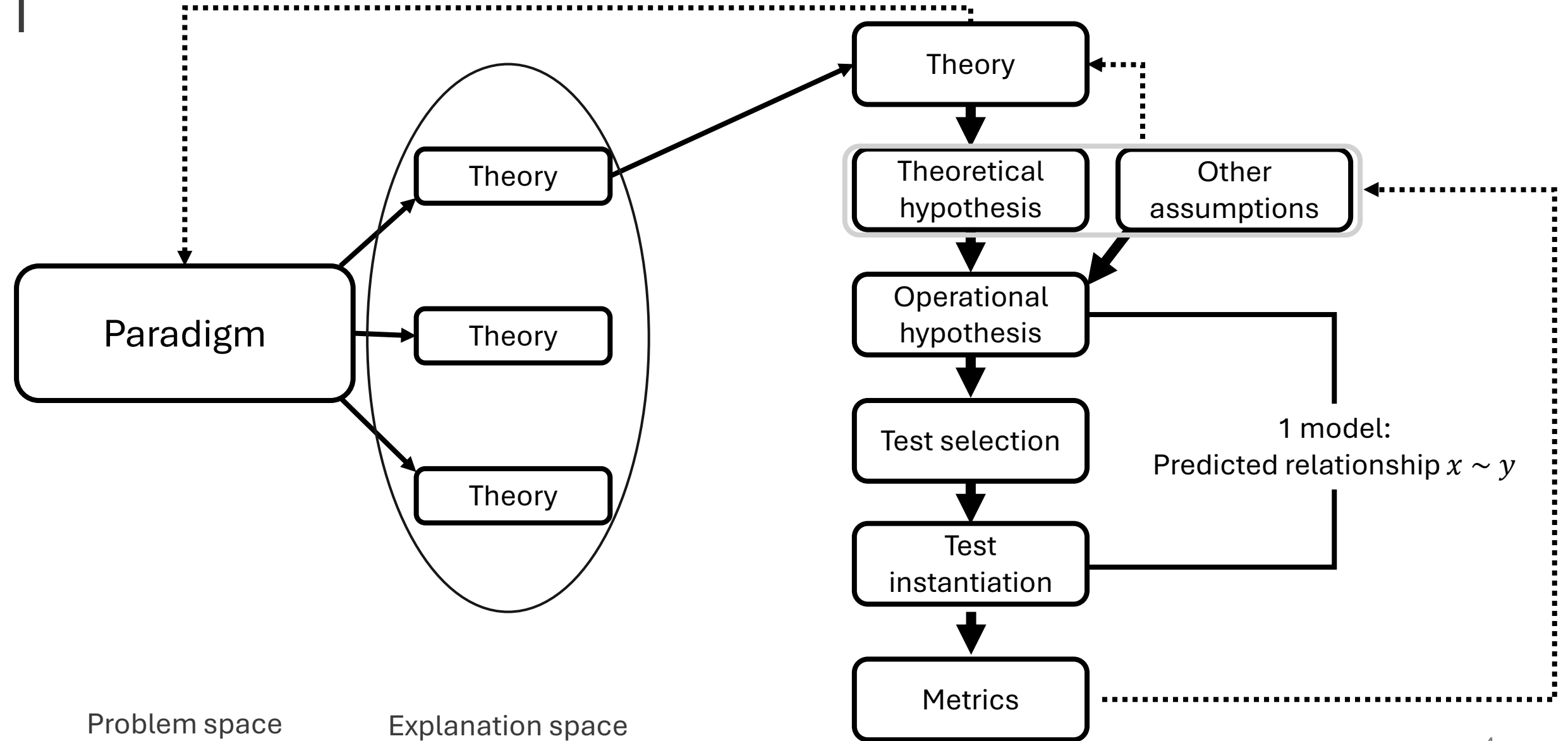
Paradigms

More specifically, a paradigm is the shared set of symbolic generalizations, models, values, and exemplars that organize how a scientific community defines, investigates, and evaluates problems.

- **Behavior:**
 - **Behaviorism** (*black box*),
 - **Cognitivism** (information-processing),
 - **Symbolism** (*steps*),
 - **Connectionism** (distributed networks),
 - **Embodied and enactive cognition**
- **Explanation:**
 - **Reductionism** (*reduced to physical*),
 - **Emergentism** (*emerges from physical*),
 - **Pluralism?** (*multiple explanatory levels*)
- **Brain:**
 - **Localizationism**,
 - **Functionalism**,
 - **Computational neuroscience** ,
 - **Dynamic systems / Networks?**
- **Any domain-level, area-level, function-level, process-level, ...**
- **Methodology** (*not Kuhnian*):
 - **Neurophysiology** (*electrophysiology, single-unit*),
 - **Neuroimaging** (*PET, fMRI, EEG, MEG*),
 - **Interventional** (*tDCS, TMS, optogenetics*)

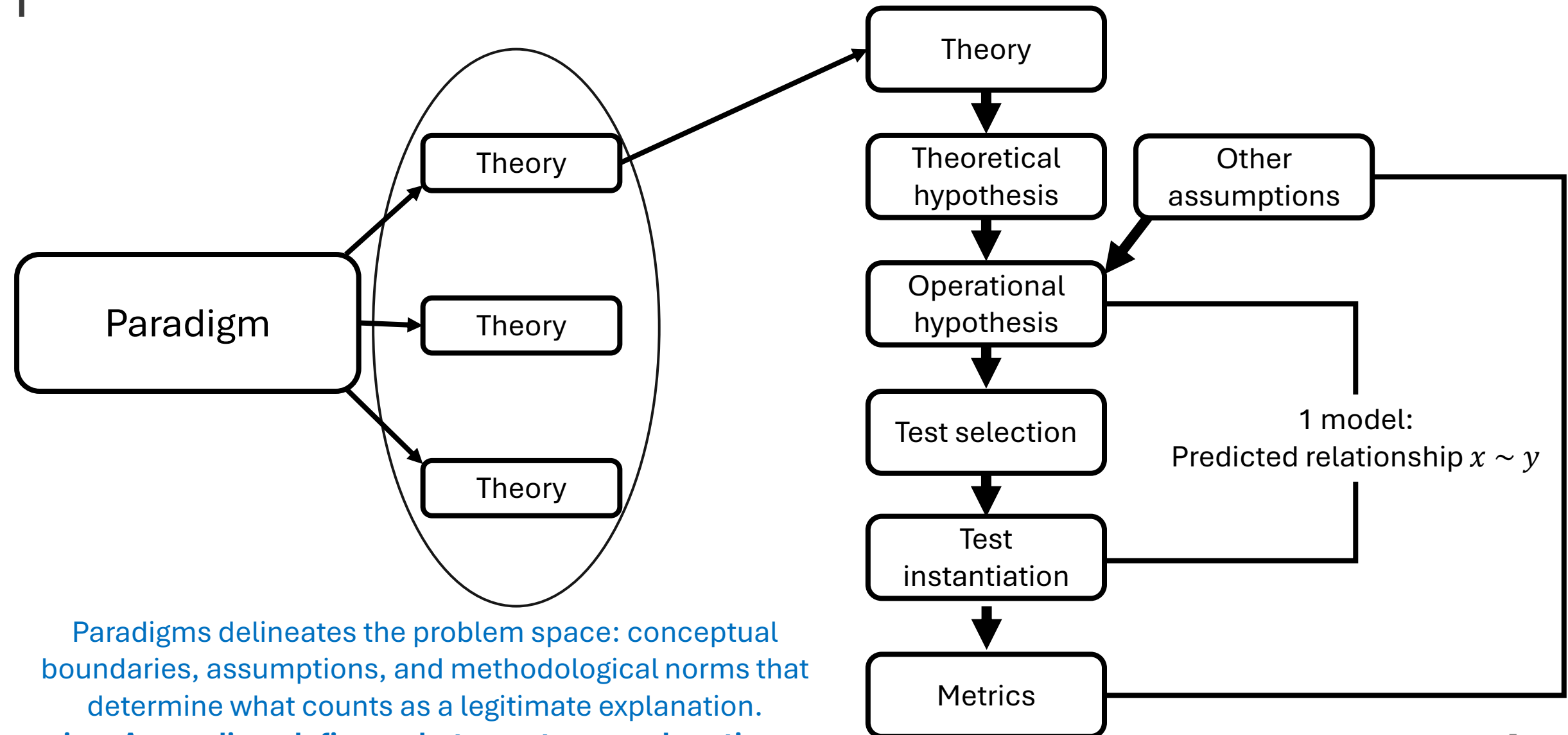
Hypothesis testing

Paradigm: shared set of symbolic generalizations, models, values, and exemplars



Hypothesis testing

Paradigm: shared set of symbolic generalizations, models, values, and exemplars



Paradigms delineates the problem space: conceptual boundaries, assumptions, and methodological norms that determine what counts as a legitimate explanation.
i.e., A paradigm defines what counts as explanation.

One cannot escape paradigms

All hypotheses have underlying assumptions.

All assumptions are grounded in a body of work (~axioms) that cohere within a given paradigm.

Accepting some assumptions automatically (& logically) rules out competing assumptions.

One cannot escape paradigms.

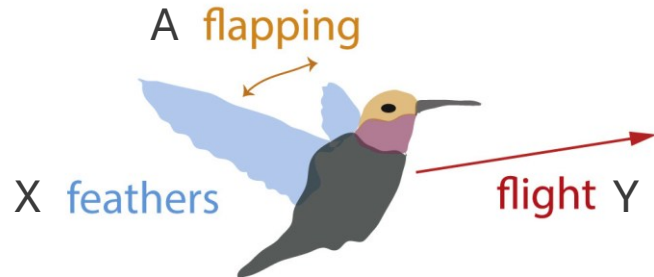
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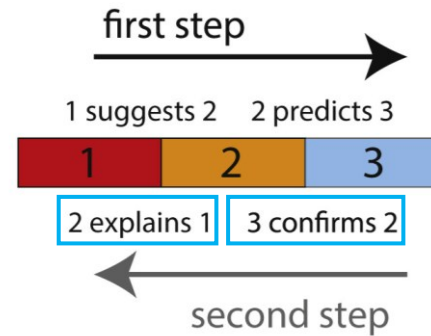
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What are theories for

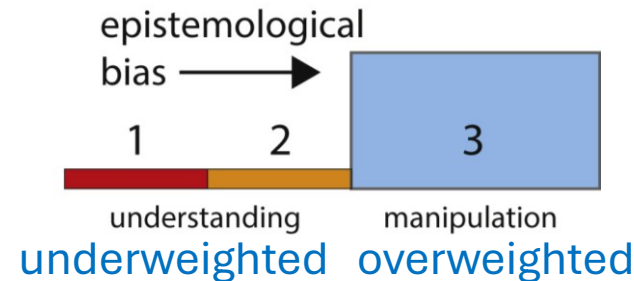
A Marr's levels of analysis



B



C



Marr:

A model only counts as explanatory if it captures a sufficient range of the processes it claims to represent.

- Describe:** We observe X, Y; X is *involved* in Y
- Explain:** Y occurs *because* the system is solving a problem by algorithm A implemented via X
- Predict:** Given state S and model M ($Y \sim X$), the system will produce $\hat{Y}$

Confirmation: A is likely because X is present

Reverse inference: Y is present because X is present

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- **Bickle, 2016. *Frontiers in Systems Neuroscience***

Neuroscience advances through “tool-driven revolutions”: moments when a new technique transforms experimental practice and thereby shifts what counts as legitimate explanation.

- **Parker, 2018. *Biology & Philosophy***

Paradigm shifts in neuroscience are primarily conceptual. Tools were necessary for empirical progress but insufficient to drive Kuhnian revolutions.

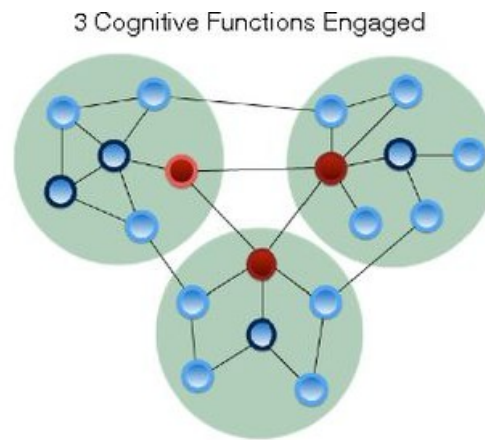
The need for qualitative leaps

Paradigms impose constraints on problem formalization and grammar

Fodor's (1983) modularity

1. Domain specificity
2. Mandatory operation
3. Limited central accessibility
4. Fast processing
5. Informational encapsulation
6. 'Shallow' outputs
7. Fixed neural architecture
8. Breakdown patterns
9. Ontogenetic pace

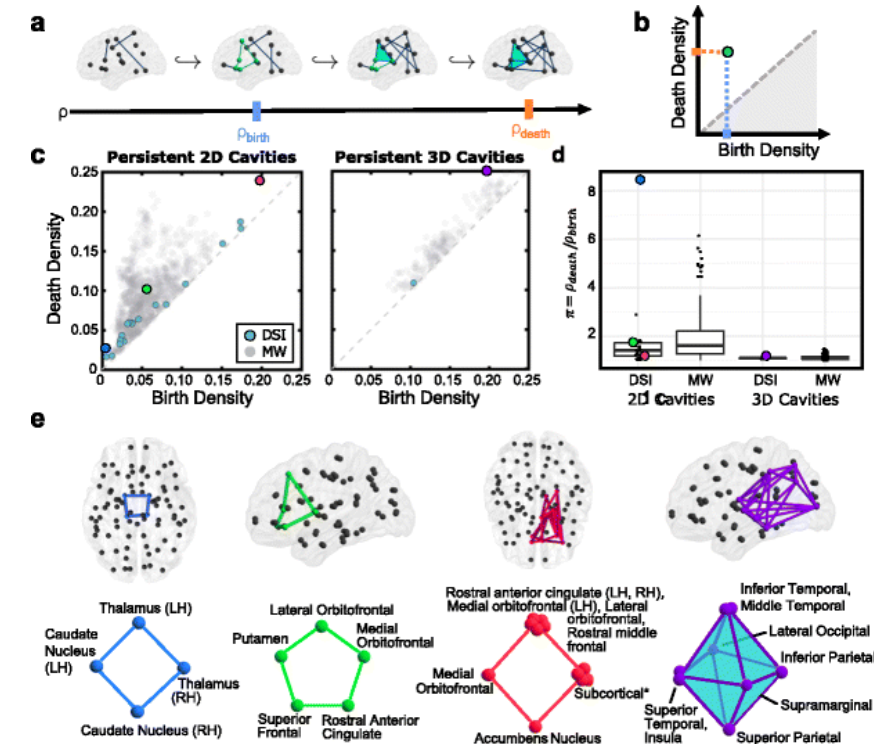
Old paradigm
Structural



Bertolero, Yeo & D'Esposito,
2015. PNAS.

Model of *modular functional
architecture*

Qualitative leap
Connectionist



Sizemore et al., 2017. *Journal of
Computational Neuroscience*
Dynamics of cliques and cavities

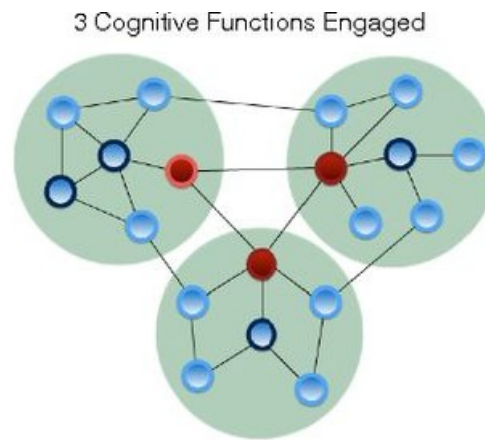
New paradigm?
Topological

The need for qualitative leaps

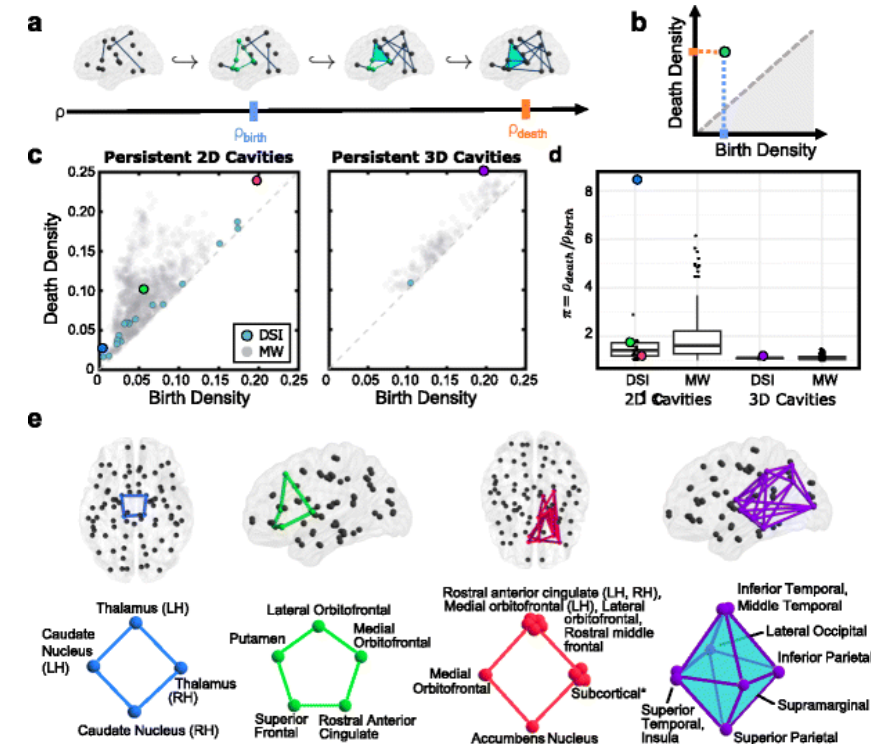
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What counts as explanation?

If the current paradigm is topological, does a structural account counts as explanation?

Implementing Occam's razor Or Flatland fallacy?

Gigerenzer & Brighton, 2009. *TICS*
Jolly & Chang, 2019. *Topics in Cognitive Science*

Parsimony can lure us toward simplistic accounts of high-dimensional problems (flatland fallacy).

- If parsimonious is preferred between many explanations (Occam's Razor, higher truthlikeness) &
- If psychological phenomenon are high-dimensional,
- How can we form an ethos for evaluating the plausibility of hypotheses that account for psychological phenomenon?
 - Are parsimonious hypotheses/models better?

When small sample sizes and high-dimensional signals, lower dimensional models with greater bias error make more accurate out-of-sample predictions.

When biases lead to better inferences
(vs. complex models)

When information is scarce, degraded, uncertain, complex, noisy
When the environment is sufficiently predictable

- **Predictive superiority**
- **Robustness to uncertainty**
Ignoring information can make predictions less sensitive to noise and small samples
- **Cognitive efficiency**
Reduces cost while maintaining sufficient performance (Martignon et al., 2008)
- **By simplifying, heuristics introduce a bias:**
 - This bias reduces the instability of predictions (variance)
 - Improves robustness and generalizability to similar situations, especially under uncertainty
- **By complexifying, models reduce bias but become more sensitive to noise :**
 - This increases the variance of predictions
 - Reduces ability to generalize to new situations



Gigerenzer & Brighton, 2009



- **We benefit from parsimony and bias** (low-dim hypotheses) **when predicting complex psychological phenomenon from small, degraded, uncertain, noisy datasets.**
- **We benefit from complexity and total evidence** (high-dim hypotheses) **when predicting complex psychological phenomenon from large datasets.**